

Cybersecurity Project Report

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Project Number: 12

Project Title: Password Hash Cracking using John the Ripper and Wordlist Customization

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1. Project Overview

Project Objectives

The primary objective of this project is to understand the mechanisms of password security and practically implement password cracking techniques using John the Ripper. This involved analyzing various hashing algorithms and evaluating the effectiveness of different attack vectors in a controlled environment.

Expected Outcomes

The anticipated learning outcomes for this project included:

- Developing a technical understanding of password hashing algorithms such as MD5, SHA, and bcrypt.
- Performing practical dictionary, brute-force, and rule-based attacks.
- Gaining proficiency in wordlist customization techniques to enhance cracking efficiency.
- Recognizing the critical importance of robust password security policies.

2. Environment and Tools Setup

Laboratory Environment

Component	Details
Operating System	Kali Linux
Virtual Machine Platform	Oracle VM VirtualBox
Network Configuration	NAT
Storage Allocated	20 GB
RAM Allocated	4 GB

Tools Selected

Tool Name	Version	Purpose
John the Ripper	Jumbo Version	Multi-purpose password cracking
RockYou Wordlist	Latest	Dictionary-based attacks
Hash Generator	md5sum (Default)	Cryptographic hash creation
Nano Editor	Default	Configuration and file editing

Tool Selection Rationale

- **John the Ripper:** A powerful, open-source tool capable of auditing hundreds of hash and cipher types.
- **RockYou Wordlist:** Contains real-world leaked passwords, providing a high success rate for dictionary attacks.
- **Kali Linux:** A standard penetration testing distribution where these security tools are pre-integrated.
- **VirtualBox:** Provides a safe, isolated sandbox for security testing.

3. Implementation Steps and Activities

Step	Activity Description	Actions/Commands	Outcome
1	Environment Setup	Install Kali Linux in VirtualBox	Successful installation
2	Tool Installation	<code>sudo apt install john</code>	Software ready for use
3	Hash Creation	Generate MD5 hashes	Initial test data prepared
4	File Creation	<code>nano hashes.txt</code>	Target file populated
5	Dictionary Attack	<code>john --wordlist=rockyou.txt</code>	Initial cracking initiated
6	Result Verification	<code>john --show</code>	Recovered passwords viewed
7	Custom Wordlist	<code>nano mywordlist.txt</code>	Targeted list created
8	Rule-based Attack	<code>john --rules</code>	Applied mangling rules
9	Brute Force	<code>john --incremental</code>	Exhaustive search performed

Key Commands Used

- **Hash Generation:** `echo -n "password123" | md5sum` (Generates the MD5 hash for a string).
- **Dictionary Attack:** `john --wordlist=/usr/share/wordlists/rockyou.txt hashes.txt` (Attempts to match hashes against the RockYou list).
- **Incremental Attack:** `john --incremental hashes.txt` (Runs a brute-force attack to find passwords by trying all combinations).

4. Evidence of Activities Performed

No.	Screenshot Reference	Description	Related Step
1	screenshot_01.png	Software Installation Confirmation	Step 2

No.	Screenshot Reference	Description	Related Step
2	screenshot_02.png	MD5 Hash Generation Process	Step 3
3	screenshot_04.png	Active Dictionary Attack on Hashes	Step 5
4	screenshot_05.png	Results Displaying Cracked Passwords	Step 6
5	screenshot_10.png	Final Summary of Project Output	Step 11

5. Troubleshooting and Problem Resolution

No.	Problem Description	Root Cause	Solution Applied
1	Tool not found	Package not installed	Executed <code>sudo apt install john</code>
2	Wordlist error	Incorrect file path	Provided full path to RockYou.txt
3	Cracking failed	Weak default wordlist	Created a customized wordlist
4	Permission denied	Lack of privileges	Preceded commands with <code>sudo</code>

6. Learning and Findings

Key Lessons Learned

- **Path Accuracy:** Precise file paths are essential for tool execution in Linux environments.
- **Wordlist Quality:** The effectiveness of a cracking attempt is directly proportional to the quality and relevance of the wordlist used.
- **Performance:** Brute-force attacks are significantly more resource-intensive and slower than dictionary-based attacks.

Project Outcomes

The project successfully demonstrated that weak passwords can be cracked in seconds using standard hardware and dictionary attacks. The use of custom wordlists and mangling rules significantly improved the success rate against passwords that included minor variations (e.g., adding "123" to a common word).

7. Conclusion

This project successfully met all learning objectives. By using John the Ripper, I gained practical experience in password auditing and the mechanics of cryptographic hashes. The most valuable takeaway is the vulnerability of simple passwords and the necessity of using complex, long-form passwords to protect sensitive systems.

8. References

1. Openwall. John the Ripper Documentation. <https://www.openwall.com/john/>
2. Kali Linux Documentation. Password Cracking Tools.
3. RockYou Wordlist Archive.

Declaration

I certify that this report represents my own work and that all sources have been properly acknowledged.

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